

Assignment 2 Solution

1, The following character encoding is used in a data link protocol: A: 01000111 B: 11100011 FLAG: 01111110 ESC: 11100000 Show the bit sequence transmitted (in binary) for the four-character frame A B ESC FLAG when each of the following framing methods is used: (a) Byte count. (b) Flag bytes with byte stuffing. (c) Starting and ending flag bytes with bit stuffing.

The solution is (a) 00000101 01000111 11100011 11100000 01111110

(b) 01111110 01000111 11100011 11100000 11100000 11100000 01111110 01111110

(c) 01111110 01000111 110100011 111000000 011111010 01111110

2, A bit string, 011110111110111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing?

The output is 011110111110011111010.

3, One of your classmates, Scrooge, has pointed out that it is wasteful to end each frame with a flag byte and then begin the next one with a second flag byte. One flag byte could do the job as well, and a byte saved is a byte earned. Do you agree?

If you could always count on an endless stream of frames, one flag byte might be enough. But what if a frame ends (with a flag byte) and there are no new frames for 15 minutes? How will the receiver know that the next byte is actually the start of a new frame and not just noise on the line? The protocol is much simpler with starting and ending flag bytes.

4, What is the maximum overhead in byte-stuffing algorithm?

The maximum overhead occurs when the payload consists of only ESC and FLAG bytes. In this case, there will be 100% overhead.

5, A channel has a bit rate of 4 kbps and a propagation delay of 20 msec. For what range of frame sizes does stop-and-wait give an efficiency of at least 50%?

Efficiency will be 50% when the time required to transmit the frame equals the round-trip propagation delay. At a transmission rate of 4 bits/msec, 160

bits takes 40 msec. For frame sizes above 160 bits, stop-and-wait is reasonably efficient.